Project 2 Module B

Objective:

The objective of Module B is to write a multi-threaded application using pthread library. To that end, we will convert the single-threaded application of Module A to multi-threaded form.

Project Description:

You are given the same input image file as in Module A. The name of input file is "input.pgm". The given image file is a greyscale image. You need to convert the image to black and white form.

This time, **you need to use multiple threads** with each thread assigned a subset of image rows. Then each thread performs the threshold operation on pixels of assigned rows.

For example, if the image has height of 512 and width of 256 then you can consider it to have 512 rows and 256 columns i.e., each row has 256 pixels.

Rows are equally divided among user-specified number of threads. For example, if user specifies 4 threads, then 4 new threads are created by the parent thread and each newly created thread is made responsible for performing threshold operation on pixels of 512/4=128 rows i.e.,

- Thread 0 performs operation on pixels of row 0 to row 127 (including row 127).
- Thread 1 performs operation on pixels of row 128 to row 255 (including row 255).
- Thread 2 performs operation on pixels of row 256 to row 383 (including row 383).
- Thread 3 performs operation on pixels of row 384 to row 511 (including row 511).

The main thread (the one the creates user-specified number of threads) waits for all other threads to completed before proceeding.

Given Files:

1. project2ModB.h
2. skeleton_project2moduleB.c
3. input.pgm
4. Makefile

Required:

1. Complete the skeleton code by writing all the missing parts (indicated by ToDos 1,2, and 3) in project2ModB.h and skelecton_project2moduleB.c

2. After completing the missing parts, run the executable file of skelecton_project2moduleA with following command where 128 is the threshold and 4 is user-specified number of threads.

   ./skelecton_project2moduleA input.pgm output.pgm 128 4

3. Draw a graph by filling the column B, C, and D (by running your program three times for a given number of thread count) of the given excel sheet.

4. Find number of cores on your computer (on Ubuntu you can do "lscpu" on terminal to find number of cores). Write 4-5 sentences to explain the graph, any increase/decrease in normalized execution time with increasing number of thread count, the relationship of the increase/decrease with number of cores.

Deliverables.

1. **Complete Source Code:** 60 points for the working code. No points for incomplete code. You must submit
   a. input image file.
   b. Make file
   c. Completed skelecton_project2moduleB.c
   d. Completed project2ModB.h

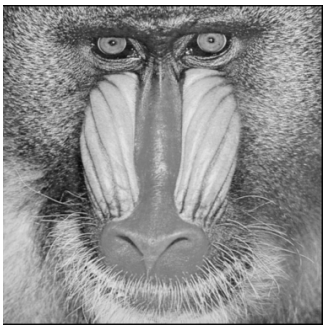
2. **Filled Excel Sheet with Graph**
   a. **Evaluation:** 20 points for drawing the graph

3. **4-5 line explanation of the graph.**
   a. **Evaluation:** 20 points for the explanation.

Submission Guidelines:

Place all your files including Makefile in a single folder, name it as first_student_id_first_student_name_second_student_id_second_student_name, zip it, and submit.

Sample Input:



Sample Output:



Sample Graph:

